AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2025

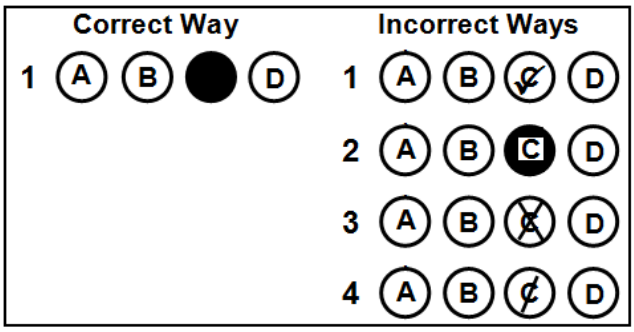
Computer Science Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. To handle the massive amounts of data and powering essential applications, the type of computer that can be used in large business organisations is

   A. minicomputer.
   B. supercomputer.
   C. microcomputer.
   D. mainframe computer.

2. There are four stages in data processing cycle and in each stage a number of tasks are performed.

   TWO tasks that are performed during processing stage are

   A. data capture and presenting data to user.
   B. executing instructions and presenting data to user.
   C. data capture and transform raw data into information.
   D. executing instructions and transform raw data into information.

3. The PRIMARY purpose of an operating system (OS) is to

   A. manage hardware resources.
   B. run applications exclusively.
   C. connect to the internet.
   D. provide security only.

4. The type of software that is typically available for free but may have limitations on features/ time of use is a/ an

   A. open-source software.
   B. licensed software.
   C. shareware.
   D. freeware.

5. The MAIN purpose of the utility software is to

   A. provide a platform for running applications and managing system resources.
   B. perform routine tasks that optimise the performance of a computer system.
   C. enhance the graphics and user interface of an operating system.
   D. compile and execute high-level programming languages.

6. The kind of application software that is made especially for data analysis and mathematical computations is

   A. word processor.
   B. web browser.
   C. spreadsheet.
   D. editor.

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7. Which of the following pointing devices is BEST suited for precise drawing and design work on a computer screen?

   A. Joystick
   B. Trackball
   C. Touchpad
   D. Graphics tablet

8. In which scenario would you use a touch screen with capacitive technology when using a device?

   A. Where precise input is required using a stylus pen.
   B. In medical equipment that require pressure-based input.
   C. Where electrical conductance of the human skin is required.
   D. In a bright outdoor environment that requires high visibility.

9. The device that is commonly used in automated teller machine (ATM)s for reading bank card information is

   A. barcode reader.
   B. optical mark reader (OMR).
   C. magnetic card reader (MCR).
   D. quick response (QR) code reader.

10. The data capture device that is typically used for automated cheque processing in banks is

    A. magnetic ink character reader (MICR).
    B. quick response (QR) code reader.
    C. optical character reader (OCR).
    D. barcode reader.

11. If one character occupies 8 bits in a computer memory, then the number of bytes required to store the word ADD-ON will be

    A. 5 bytes.
    B. 6 bytes.
    C. 40 bytes.
    D. 48 bytes.

12. How many gibibytes (GiB) are in 2 tebibyte (TiB)?

    A. 2048 GiB
    B. 2048 MiB
    C. 1024 MiB
    D. 1024 GiB

13. The statement that describes the read only memory (ROM) is that it is used

    A. for temporary storage of data.
    B. to store firmware and is non-volatile.
    C. to modify existing data stored in ROM.
    D. to add new or delete previous data stored in ROM.

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14. The PRIMARY function of the cache memory in a processor is to

A. store frequently used data.
B. store user files permanently.
C. store the operating system (OS).
D. provide additional storage space.

15. Which of the following characteristics is TRUE about Dynamic Random Access Memory (DRAM)?

A. It does not require refreshing.
B. It uses a single transistor per cell.
C. It is faster than static RAM (SRAM).
D. It does not require electricity to retain data.

16. Which type of Random Access Memory (RAM) is generally used for Central Processing Unit (CPU) cache?

A. SRAM
B. DRAM
C. SDRAM
D. DDR SDRAM

17. How does the size and speed of L3 cache generally compare to L1 and L2 caches?

A. L3 cache is larger and slower than L1 and L2 caches.
B. L3 cache is smaller and faster than L1 and L2 caches.
C. L3 cache is the same size and speed as L1 and L2 caches.
D. L3 cache is the same size but faster than L1 and L2 caches.

18. Following are the devices used for storing data.

I. CD
II. DVD
III. Blu-Ray

The device(s) that use(s) a red-violet laser to read and write data is/ are

A. I only.
B. III only.
C. I and II.
D. II and III.

19. Flash memory and Secure Digital (SD) cards store data by using

A. electrons in a memory cell made up of transistors.
B. a magnetic coating on a spinning disk.
C. a laser to read and write data.
D. temporary storage method.

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20. The primary function of a register in a CPU is to

A. manage data flow.
B. store long-term data.
C. hold data temporarily for processing.
D. connect external devices through bus.

21. The register that is used to store intermediate results of calculations in the CPU is

A. Stack Pointer (SP).
B. Accumulator (AC).
C. Base Register (BR).
D. Status Register (SR).

22. In the execute phase of the fetch-decode-execute cycle, the role of CPU is to

A. retrieve the next instruction from memory.
B. store the result back in the memory.
C. interpret the instruction.
D. decode the instruction.

23. All of the following are the components of the CPU EXCEPT

A. registers.
B. system unit.
C. control unit.
D. internal buses.

24. The examples of general-purpose registers are the

A. base register and counter register.
B. data register and instruction register.
C. instruction register and accumulator register.
D. accumulator register and memory address register.

25. Data transfer instruction is used to transfer data between the

A. RAM and I/O.
B. I/O and hard disk.
C. register and RAM.
D. register and hard disk.

26. The port that is used for transmitting high-definition video and audio signals from a computer to a display is

A. HDMI port.
B. Serial port.
C. USB port.
D. PS/2 port.

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27. The card used to connect a computer to a wired network, providing high-speed network access is a

A. wireless network card
B. modem card
C. gigabit card
D. sound card

28. The number of channel paths used by a Single Inline Memory Module (SIMM) and Dual Inline Memory Module (DIMM) are

A. SIMMs have a 64-bit data path, while DIMMs have a 32-bit data path.
B. SIMMs have a 32-bit data path, while DIMMs have a 64-bit data path.
C. SIMMs and DIMMs both have 32-bit data paths.
D. SIMMs and DIMMs both have 64-bit data paths.

29. The THREE main components in a system unit are

A. casing, mouse and monitor.
B. monitor, keyboard and mouse.
C. casing, power supply and motherboard.
D. keyboard, power supply and motherboard.

30. The PRIMARY purpose of a computer network is to

A. allow communication and resource sharing.
B. process and store complex calculations.
C. store large amounts of data.
D. enhance graphic rendering.

31. Read the following features.

- Communication within a few centimetres to a few metres
- Commonly used for devices such as wearable technology

The type of network that is described in the given features is

A. Metropolitan Area Network (MAN).
B. Interplanetary Internet (IPN).
C. Wide Area Network (WAN).
D. Body Area Network (BAN).

32. Three Open System Interconnection (OSI) model layers are merged as one layer in Transmission Control Protocol/ Internet Protocol (TCP/ IP) model, i.e. Application layer.

If one of these layers is application layer, the other TWO OSI model layers would be

A. physical and network.
B. network and transport.
C. session and presentation.
D. transport and presentation.

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33. The topology in which each node interconnected with many or all other nodes, providing multiple paths for data transmission is

A. bus.
B. star.
C. ring.
D. mesh.

34. According to which network topology, every node forms a closed loop with a single point-to-point connection?

A. Star
B. Ring
C. Tree
D. Mesh

35. The wireless technology that is commonly used for short-range communication between devices, often within the range of about 10 metres, is

A. LTE.
B. Wi-Fi.
C. WiMAX.
D. Bluetooth.

36. The network in which users can be added without changing the physical connections of the network is a/ an

A. infra-red network.
B. optic fibre network.
C. coaxial cable network.
D. telephone line network.

37. The term 'line of sight' in the context of wireless communication refers to the

A. direct visual path between the transmitter and receiver.
B. curvature of the Earth affecting signal transmission.
C. distance a signal can travel without losing strength.
D. frequency of the radio signal.

38. The mobile communication technology that allows simultaneous voice and data transmission on the same channel is known as

A. GSM (Global System for Mobile communication).
B. FDMA (Frequency Division Multiple Access).
C. CDMA (Code Division Multiple Access).
D. LTE (Long Term Evolution).

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39. The PRIMARY use of Global Positioning System (GPS) technology is to

A. enhance computer networks.
B. speed up internet connection.
C. provide real-time navigation.
D. share large amounts of data through satellite.

40. Which of the following is the disadvantage of using a Database Management System (DBMS)?

A. Data inconsistency
B. Limited data access
C. Difficult data integration
D. High implementation cost

41. Consider the following characteristics of a database:

- Tree like structure
- Single parent
- Potentially multiple children

The database model based on the given characteristics is called

A. network model.
B. hierarchical model.
C. object-oriented model.
D. object-relational model.

42. In the context of relational model, the foreign key is used to

A. uniquely identify each record in a table.
B. enforce data type constraints.
C. link two tables together.
D. store large text data.

43. A record in a database is defined as a

A. single piece of data.
B. collection of related fields.
C. visual representation of data.
D. storage format for digital data.

44. In a database table, another term used for a column is

A. record.
B. tuple.
C. field.
D. file.

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45. An employee database table has following fields. The Employee_ID is the primary key.

I. Employee_ID
II. Employee_Name
III. Employee_National_ID_Card_Number
IV. Employee_Age
V. Employee_Gender
VI. Employee_Address

In the given options, the MOST appropriate table field having number data type is

A. II only.
B. IV only.
C. V only.
D. VI only.

46. Which MS Access object allows end users to create a customised interface for data entry and editing?

A. Table
B. Form
C. Query
D. Report

47. In MS-Access, the PRIMARY function of a table is to

A. filter data based on specific criteria.
B. store data in rows and columns.
C. generate graphical reports.
D. view data on screen.

48. If I want to remove a table from database in MS-Access, the data definition language command in SQL would be

A. TRUNCATE TABLE
B. REMOVE TABLE
C. DELETE TABLE
D. DROP TABLE

49. The MS Access query that will fetch records that consist of CarModel and CarPrice fields from a database table named as Cars is

A. Select CarModel, CarPrice From Cars;
B. Select *CarModel, *CarPrice From Cars;
C. Select CarModel AND CarPrice From Cars;
D. Select *(CarModel AND CarPrice) From Cars;

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50. The primary purpose of the WHERE statement in SQL within MS-Access is to

A. update existing data in a table.
B. insert new data into a table.
C. filter data from a table.
D. delete data from a table.

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END OF PAPER

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